Université d'Ottawa · University of Ottawa
Faculté des sciences Mathématiques et de statistique
Faculty of Science Mathematics and Statistics

Introduction to Linear Algebra MAT 1341A

Midterm 2 - Version A

November 16, 2023

You must **sign below** to confirm that you have read, understand, and will follow these **instructions**:

- This is an 75-minute **closed-book** exam; no notes are allowed. **Calculators are not permitted.** This test has been designed so that a calculator will not be needed.
- The exam consists of 6 questions on 8 pages. Page 8 provides additional work space and you may also use the backs of any of the pages if needed. Indicate clearly where your answer can be found if it is not written down immediately after where the question is given.
- Question 1–3 are short-answer questions. No justification is required for these questions.
- Questions 4–6 are long-answer questions. You must show all relevant steps and give appropriate justifications in order to obtain full marks.
- On Page 7, there is a bonus question. Only attempt it after you have finished all the other questions.
- **Cellular phones** and other electronic devices **are not permitted** during this exam. Phones and other devices must be turned off completely and stored out of reach. Do not keep them in your possession, such as in your pockets. If you are caught with such a device, the following may occur: academic fraud allegations will be filed which may result in your obtaining a 0 (zero) for the exam.

LAST NAME: _____

First name: _____

Student Number: _____

Signature: _____

Please circle the DGD section where you would like to have your test returned to you:

DGD 1	DGD 2	DGD 3	DGD 4
MON 1:00 – 2:20	MON 1:00 – 2:20	MON 5:30 – 6:50	MON 4:00 – 5:20
STE C0136	MRT 221	MRT 256	MRT 256

Quest. 1	Quest. 2	Quest. 3	Quest. 4	Quest. 5	Quest. 6	Bonus	Total
/12	/5	/5	/8	/10	/10	/3	/50

(1) Let A be a 2023×2021 matrix. Which of the following statements are true? Which are false?

C1. $\text{Col}(A)$ is a subspace of $\mathbb{R}^{2021}$.

C2. If the dimension of $\text{Col}(A)$ is 1000, then the dimension of $\text{Null}(A)$ is 1021.

C3. If the dimension of $\text{Row}(A)$ is 1000, then the dimension of $\text{Null}(A)$ is 1023.

C4. The dimension of $\text{Null}(A)$ is always strictly greater than zero.

C5. If $\mathbf{b}$ is an element of $\text{Col}(A)$, then the equation $A\mathbf{x} = \mathbf{b}$ has a solution.

C6. The set of solutions to the equation $A\mathbf{x} = \mathbf{b}$ is a subspace of $\mathbb{R}^{2021}$ for all $\mathbf{b} \in \mathbb{R}^{2023}$.

In each box below, circle T if the statement is true or F if the statement if else. [12]

C1	C2	C3	C4	C5	C6
T / F	T / F	T / F	T / F	T / F	T / F

(2) Let $\mathbb{P}_2$ be the set of polynomials of degree at most 2. Which of the following sets are bases of $U = \{f(X) \in \mathbb{P}_2 : f(X) = f(-X)\}$? There may be more than one correct answer.

A. $\{X^2 + 1, X^2 - 2\}$.

B. $\{X^2 - 1, 1, X^2 + 4\}$.

C. $\{X^2, 1\}$.

D. $\{X^2, 3\}$.

E. $\{1, X, X^2\}$.

Answer: A, C, D [5]

(3) Let $\mathbf{v}_1 = \mathbf{0}$, $\mathbf{v}_2 = (1, 1, 0)$, $\mathbf{v}_3 = (2, 1, 1)$, $\mathbf{v}_4 = (1, 0, 0)$, $\mathbf{v}_5 = (0, 1, 1)$. Which of the following sets are linearly independent? There may be more than one correct answer.

A. $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$.

B. $\{\mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$.

C. $\{\mathbf{v}_3, \mathbf{v}_4, \mathbf{v}_5\}$.

D. $\{\mathbf{v}_2, \mathbf{v}_4, \mathbf{v}_5\}$.

E. $\{\mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_5\}$.

Answer: B, D, E [5]

(4) Let

$$A = \begin{bmatrix} 1 & 3 & 2 \\ 1 & 4 & 4 \\ -1 & -1 & 1 \end{bmatrix}.$$

Is A invertible? If it is, find A^{-1}. [8]

$$\left[\begin{array}{ccc|ccc} 1 & 3 & 2 & 1 & 0 & 0 \\ 1 & 4 & 4 & 0 & 1 & 0 \\ -1 & -1 & 1 & 0 & 0 & 1 \end{array}\right]$$

$-R_1+R_2\to R_2$, $R_1+R_3\to R_3$ $\sim$

$$\left[\begin{array}{ccc|ccc} 1 & 3 & 2 & 1 & 0 & 0 \\ 0 & 1 & 2 & -1 & 1 & 0 \\ 0 & 2 & 3 & 1 & 0 & 1 \end{array}\right]$$

$-3R_2+R_1\to R_1$, $-2R_2+R_3\to R_3$ $\sim$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -4 & 4 & -3 & 0 \\ 0 & 1 & 2 & -1 & 1 & 0 \\ 0 & 0 & -1 & 3 & -2 & 1 \end{array}\right]$$

$-R_3\to R_3$ $\sim$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -4 & 4 & -3 & 0 \\ 0 & 1 & 2 & -1 & 1 & 0 \\ 0 & 0 & 1 & -3 & 2 & -1 \end{array}\right]$$

$4R_3+R_1\to R_1$, $-2R_3+R_2\to R_2$ $\sim$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -8 & 5 & -4 \\ 0 & 1 & 0 & 5 & -3 & 2 \\ 0 & 0 & 1 & -3 & 2 & -1 \end{array}\right]$$

So A is invertible, $A^{-1} = \begin{bmatrix} -8 & 5 & -4 \\ 5 & -3 & 2 \\ -3 & 2 & -1 \end{bmatrix}$

(5) Let $\mathbf{v}_1 = \begin{bmatrix}1\\1\\0\\5\end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix}0\\1\\1\\2\end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix}1\\0\\-1\\3\end{bmatrix}$.

(a) Find a matrix A so that Null(A) =Span$\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$. [6]

$$\left[\begin{array}{ccc|c}1&0&1&x_1\\1&1&0&x_2\\0&1&-1&x_3\\5&2&3&x_4\end{array}\right] \xrightarrow[-5R_1+R_4\to R_4]{-R_1+R_2\to R_2} \left[\begin{array}{ccc|c}1&0&1&x_1\\0&1&-1&x_2-x_1\\0&1&-1&x_3\\0&2&-2&x_4-5x_1\end{array}\right]$$

$$\xrightarrow[-2R_2+R_4\to R_4]{-R_2+R_3\to R_3} \left[\begin{array}{ccc|c}1&0&1&x_1\\0&1&-1&x_2-x_1\\0&0&0&x_3-x_2+x_1\\0&0&0&x_4-3x_1-2x_2\end{array}\right]$$

So $\begin{bmatrix}x_1\\x_2\\x_3\\x_4\end{bmatrix} \in \text{Span}\{v_1, v_2, v_3\} \iff$ $x_3 - x_2 + x_1 = 0$, $x_4 - 3x_1 - 2x_2 = 0$

$\iff \begin{bmatrix}x_1\\x_2\\x_3\\x_4\end{bmatrix} = \text{Null}(A)$, where $A = \begin{bmatrix}1&-1&1&0\\3&-2&0&1\end{bmatrix}$

(b) Find a basis of Null(B), where $B = \begin{bmatrix}1&0&1\\1&1&0\\0&1&-1\\5&2&3\end{bmatrix}$. [4]

Calculations in (a) show $B \sim \begin{bmatrix}1&0&1\\0&1&-1\\0&0&0\\0&0&0\end{bmatrix}$

So $\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix} \in \text{Null}(B) \iff x_1 = -x_3,\ x_2 = x_3,\ x_3 = s \in \mathbb{R}$

So $\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix} = s\begin{bmatrix}-1\\1\\1\end{bmatrix}$. $\left\{\begin{bmatrix}-1\\1\\1\end{bmatrix}\right\}$ is a basis of Null(B).

(6) Let $\mathbf{v}_1 = \begin{bmatrix} 3 \\ 1 \\ -1 \\ 2 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 0 \\ -1 \\ 1 \\ 2 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 3 \\ 0 \\ 0 \\ 4 \end{bmatrix}$.

(a) Find a basis of Span$\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$. [5]

$$\begin{bmatrix} 3 & 0 & 3 \\ 1 & -1 & 0 \\ -1 & 1 & 0 \\ 2 & 2 & 4 \end{bmatrix} \xrightarrow{\frac{1}{3}R_1 \to R_1} \begin{bmatrix} 1 & 0 & 1 \\ 1 & -1 & 0 \\ -1 & 1 & 0 \\ 2 & 2 & 4 \end{bmatrix}$$

$$\begin{matrix} -R_1 + R_2 \to R_2 \\ R_1 + R_3 \to R_3 \\ -2R_1 + R_4 \to R_4 \end{matrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & -1 & -1 \\ 0 & 1 & 1 \\ 0 & 2 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Pivots are in columns 1 and 2,

so $\{v_1, v_2\}$ is a basis.

(b) Extend the basis you found in (a) to a basis of $\mathbb{R}^4$. [5]

Consider $\begin{bmatrix} 3 & 1 & -1 & 2 \\ 0 & -1 & 1 & 2 \end{bmatrix}$

we have pivots in columns 1 and 2.

So we can add $[0\ 0\ 1\ 0]$, $[0\ 0\ 0\ 1]$

to obtain a matrix of rank 4.

so $\left\{ v_1, v_2, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$ is

a basis of $\mathbb{R}^4$.

Bonus: Let U be a vector space of dimension 3. Prove that if $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is a spanning set of U, then it is a basis of U. [3]

Since $\{u_1, u_2, u_3\}$ is a spanning set, we need to show it's LI.

Suppose it's LD, then there is a redundant element, let's say u_1 s.t. $u_1 \in \text{span}\{u_2, u_3\}$.

Then $\text{span}\{u_1, u_2, u_3\} = \text{span}\{u_2, u_3\}$.

In a vector space,

spanning element $\geq$ dimension.

$2 \geq 3$ contradiction.